# Chen Liu

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## Education

**Yale University,** PhD in Computer Science — *New Haven, CT.* *Aug 2022 - May 2027*

- Deep Learning, Computer Vision, Neural Network Manifold, Representation Learning, Medical Imaging.
- Teaching Fellow for (1) Deep Learning on Graphs (Rex Ying), (2) AI Foundation Models (LLM) (Arman Cohan).

**Columbia University,** MS in Electrical Engineering — *New York, NY.* *Aug 2018 - Feb 2020*

- Nikola Tesla Electrical Engineering Scholar ("to the most exceptional applicants", ≈ top 10% among those admitted)

**Bucknell University,** BS in Electrical Engineering, Minor in Biomedical Engineering — *Lewisburg, PA.* *Aug 2014 - May 2018*

- Tau Beta Pi Honor, Alpha Lambda Delta Honor, ILTM student consultant

## Skills

**Research** Machine Learning, Deep Learning, Information Theory, Computer Vision, Medical Imaging (radiology): MRI, CT, etc.

**Programming** Python (PyTorch, TensorFlow, Numpy, etc.), LaTeX, Linux Bash, Git, Docker, Rust, C++

## Work Experience

**Senior Research Scientist @ GE Healthcare** — *San Ramon, CA.* *Aug 2021 - Jul 2022*

Research and development of deep learning solutions in medical imaging.

- Anatomic landmark detection in X-ray images for quantification. — Publication 5, Patent 1
- External object classification in X-ray images. — Patent 2

**Research Software Engineer @ Matician Inc** — *Palo Alto, CA.* *Jan 2021 – Jun 2021*

Developing SLAM from scratch in Rust, aiming at 30 times faster than the SOTA ORB-SLAM using only visual input.

- Re-localization and Loop Closure. Location detection, loop detection, geometric and temporal check, etc.

**Research Assistant (Funded by Grant) @ Columbia University Medical Center** — *New York, NY.* *Dec 2019 – Nov 2020*

Return offer after working in the lab. Led or participated in projects and mentored master students on research. Authored 10+ publications in conferences or journals.

- Synthesize visual effects from common but harmful contrast agents in MRI. — Publication 6, 8, 12
- Software (in MATLAB) for SOTA magnetic resonance spectroscopy processing. — Publication 9, 10, 11
- Improving dense cell segmentation with edge feature enhancement. — Publication 7

## Achievements and Services

2022 **Outstanding Reviewer Award,** International Conference on Machine Learning (ICML) — *top 10%*

– **Reviewing Committee Member,** NeurIPS 2021-23, ICLR 2022-24, ICML 2022

– **Journal Reviewer,** IEEE Trans. NNLS 2021-Present

## Selected Publications and Patents

1. Co-inventor. "System and Method for Obtaining Accurate Measurements and Quantification of X-Ray Image from Estimation of Key Anatomical Locations". ***US Patent App.*** GE Healthcare. [Patent]
2. Co-inventor. "X-Ray Lead Marker Detection System for X-Ray Imaging System". ***US Patent App.*** GE Healthcare. [Patent]
3. Danqi Liao*, Chen Liu*, et al. "Assessing Neural Network Representations During Training Using Noise-resilient Diffusion Spectral Entropy". Presented at ***ICML 2023 Workshop***. Under review at *AISTATS 2024*. [Git]
4. Chen Liu*, Matthew Amodio*, et al. "CUTS: A Framework for Multigranular Unsupervised Medical Image Segmentation". Under review at *Medical Image Analysis*. [PDF] [Git]
5. Chen Liu, et al. "Adversarial focal loss: Asking your discriminator for hard examples". *Preprint*. [PDF]
6. Chen Liu*, et al. "Deep learning of MRI contrast enhancement for mapping cerebral blood volume from single-modal non-contrast scans of aging and Alzheimer's disease brains". *Frontiers on Aging Neuroscience* (Impact Factor: 5.7). [PDF] [Git]
7. Nanyan Zhu*, Chen Liu*, et al. "Segmentation with Residual Attention U-Net and an Edge-Enhancement Approach Preserves Cell Shape Features". *IEEE EMBC 2022*. [PDF] [Git]
8. Nanyan Zhu*, Chen Liu*, et al. "Deep Learning Identifies Neuroimaging Signatures of Alzheimer's Disease Using Structural and Synthesized Functional MRI Data". *IEEE ISBI 2021*. [PDF]
9. Chen Liu, et al. "JET – A MATLAB Toolkit for Automated J-Difference-Edited MR Spectra Processing of in vivo Mouse MEGA-PRESS Study at 9.4 T". *ISMRM 2021*. [PDF] [Git]
10. Co-author. "In vivo $\gamma$-aminobutyric acid increase as a biomarker of the epileptogenic zone: An unbiased metabolomics approach". *Epilepsia* (Impact Factor: 6.7). [PDF] [Journal Commentary]
11. Co-author. "Reduced hippocampal GABA+ is associated with poorer episodic memory in healthy older women: a pilot study". *Frontiers in Behavioral Neuroscience* (Impact Factor: 3.6). [PDF]
12. Co-author. "Substituting gadolinium in brain MRI using DeepContrast". *IEEE ISBI 2020*. Best paper finalist. [PDF]